

SUPERSTORE EXPLORATORY DATA ANALYSIS

```
In [1]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

print("Libraries loaded successfully")
```

Matplotlib is building the font cache; this may take a moment.
Libraries loaded successfully

```
In [2]: df = pd.read_csv(r"C:\Users\Aleeyarh\Downloads\Superstore.csv")
df.head()
```

Out[2]:

	Category	City	Country	Customer.ID	Customer.Name	Discount	Market	记录数	C
0	Office Supplies	Los Angeles	United States	LS-172304	Lycoris Saunders	0.0	US	1	200
1	Office Supplies	Los Angeles	United States	MV-174854	Mark Van Huff	0.0	US	1	200
2	Office Supplies	Los Angeles	United States	CS-121304	Chad Sievert	0.0	US	1	200
3	Office Supplies	Los Angeles	United States	CS-121304	Chad Sievert	0.0	US	1	200
4	Office Supplies	Los Angeles	United States	AP-109154	Arthur Pritchep	0.0	US	1	200

5 rows × 27 columns



```
In [3]: df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 51290 entries, 0 to 51289
Data columns (total 27 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Category              51290 non-null  object
1   City                  51290 non-null  object
2   Country               51290 non-null  object
3   Customer.ID           51290 non-null  object
4   Customer.Name         51290 non-null  object
5   Discount              51290 non-null  float64
6   Market                51290 non-null  object
7   记录数                51290 non-null  int64
8   Order.Date            51290 non-null  object
9   Order.ID              51290 non-null  object
10  Order.Priority        51290 non-null  object
11  Product.ID            51290 non-null  object
12  Product.Name          51290 non-null  object
13  Profit                51290 non-null  float64
14  Quantity              51290 non-null  int64
15  Region                51290 non-null  object
16  Row.ID                51290 non-null  int64
17  Sales                 51290 non-null  int64
18  Segment               51290 non-null  object
19  Ship.Date             51290 non-null  object
20  Ship.Mode             51290 non-null  object
21  Shipping.Cost         51290 non-null  float64
22  State                 51290 non-null  object
23  Sub.Category          51290 non-null  object
24  Year                  51290 non-null  int64
25  Market2               51290 non-null  object
26  weeknum               51290 non-null  int64
dtypes: float64(3), int64(6), object(18)
memory usage: 10.6+ MB

```

In [4]: `df.describe()`

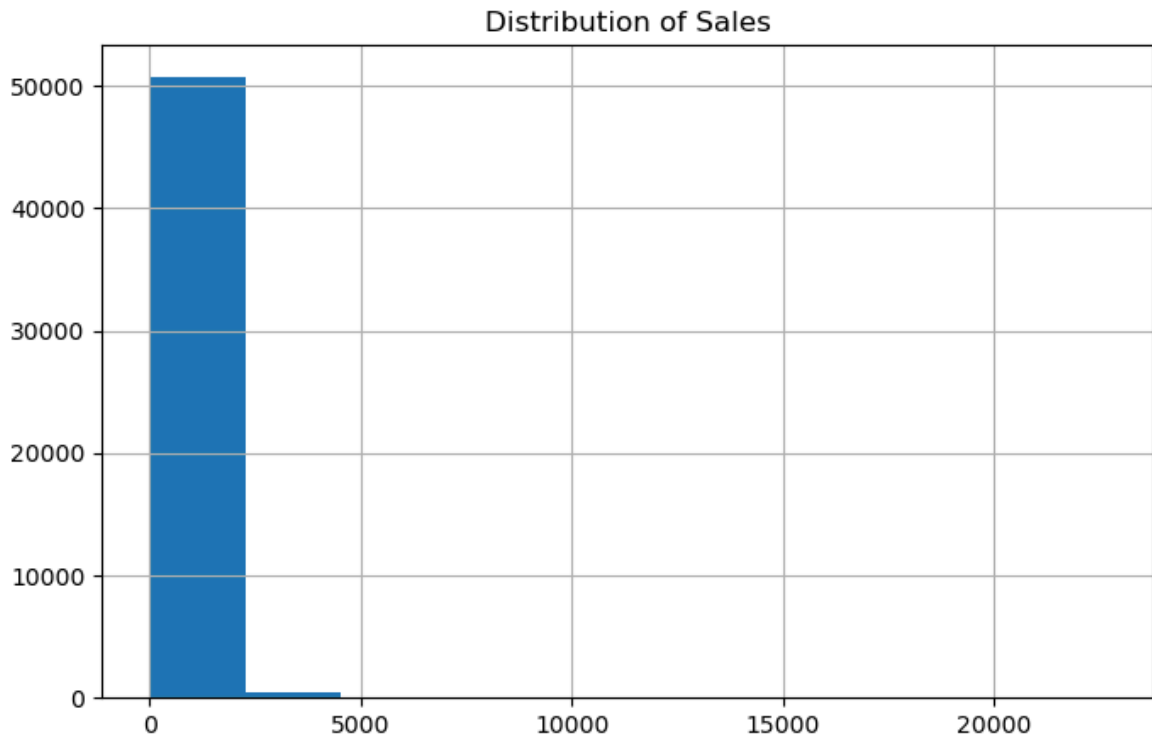
Out[4]:

	Discount	记录数	Profit	Quantity	Row.ID	Sales	S
count	51290.000000	51290.0	51290.000000	51290.000000	51290.000000	51290.000000	
mean	0.142908	1.0	28.610982	3.476545	25645.500000	246.498440	
std	0.212280	0.0	174.340972	2.278766	14806.29199	487.567175	
min	0.000000	1.0	-6599.978000	1.000000	1.000000	0.000000	
25%	0.000000	1.0	0.000000	2.000000	12823.25000	31.000000	
50%	0.000000	1.0	9.240000	3.000000	25645.50000	85.000000	
75%	0.200000	1.0	36.810000	5.000000	38467.75000	251.000000	
max	0.850000	1.0	8399.976000	14.000000	51290.00000	22638.000000	

In [5]: `df['Category'].value_counts()`

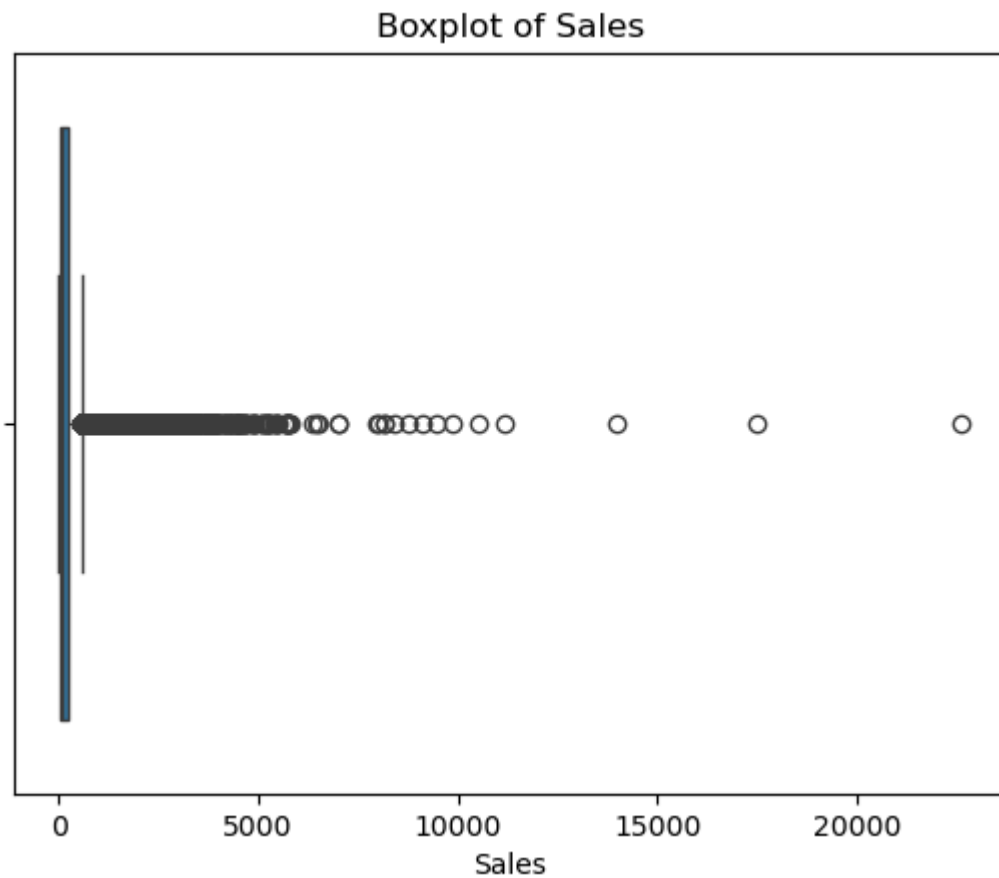
```
Out[5]: Category
Office Supplies    31273
Technology         10141
Furniture          9876
Name: count, dtype: int64
```

```
In [11]: df['Sales'].hist(figsize=(8,5))
plt.title('Distribution of Sales')
plt.show()
```



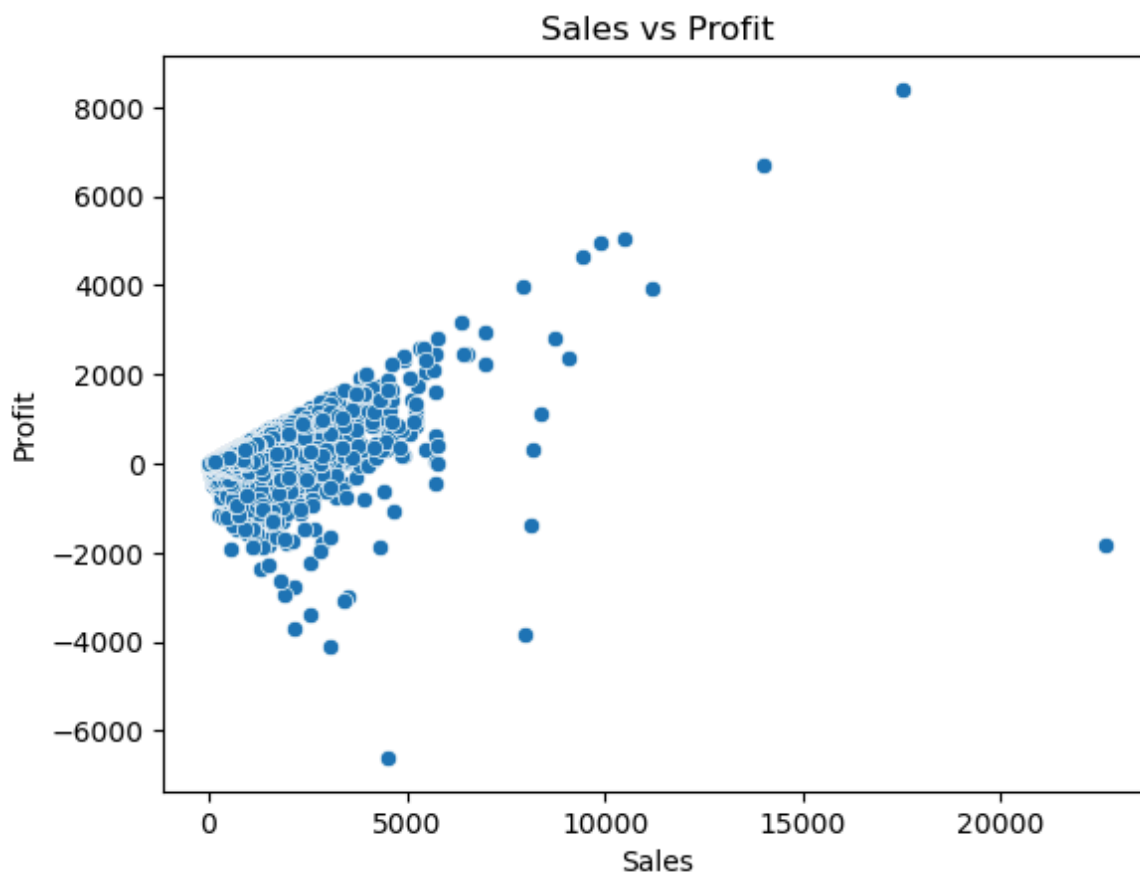
Most sales transactions are concentrated at lower values, while a few transactions have very high sales.

```
In [7]: sns.boxplot(x=df['Sales'])
plt.title('Boxplot of Sales')
plt.show()
```



The boxplot shows several outliers, indicating some sales transaction are much larger than the majority

```
In [8]: sns.scatterplot(x='Sales', y='Profit', data=df)
plt.title('Sales vs Profit')
plt.show()
```

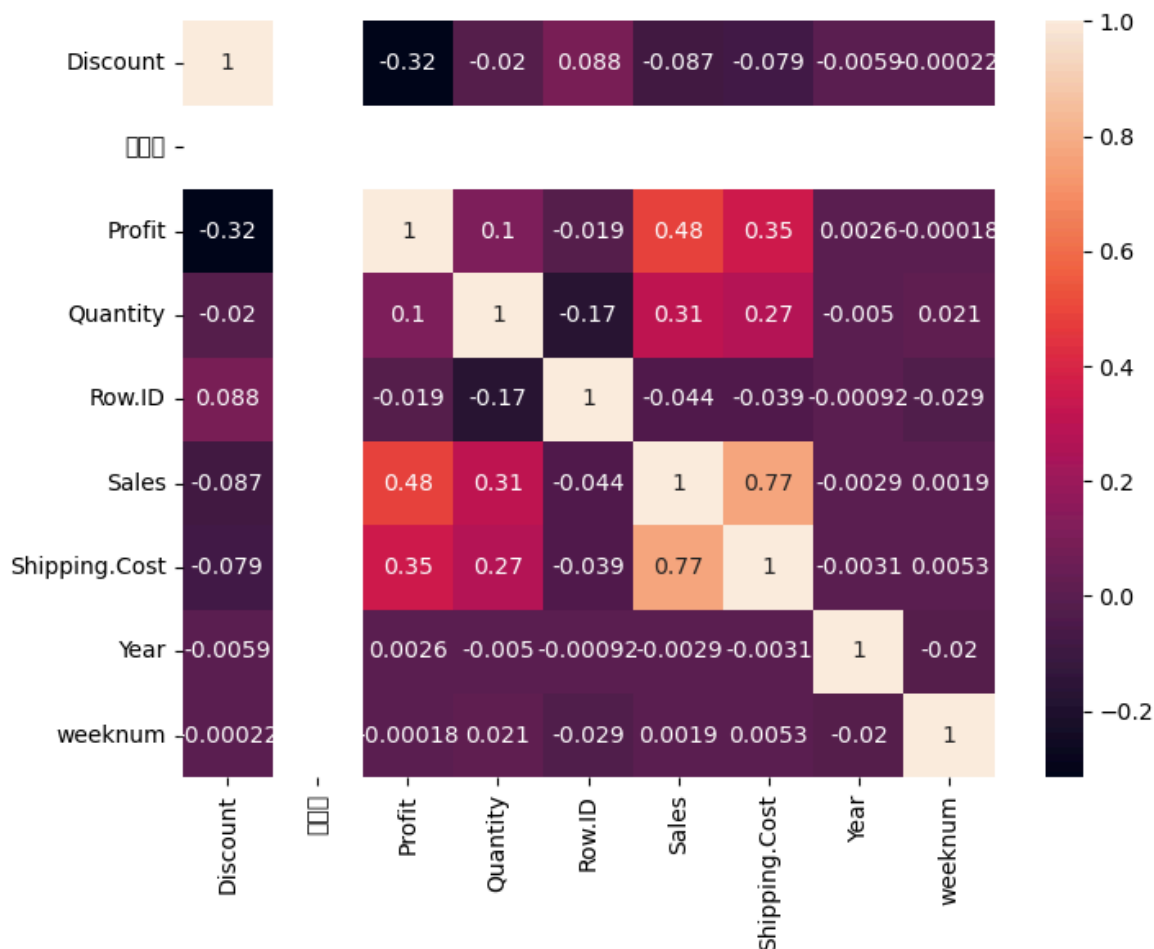


There is generally a positive relationship between sales and profit, although some high sales transactions generated low profit

```
In [9]: numeric_df = df.select_dtypes(include='number')

plt.figure(figsize=(8,6))
sns.heatmap(numeric_df.corr(), annot=True)
plt.show()
```

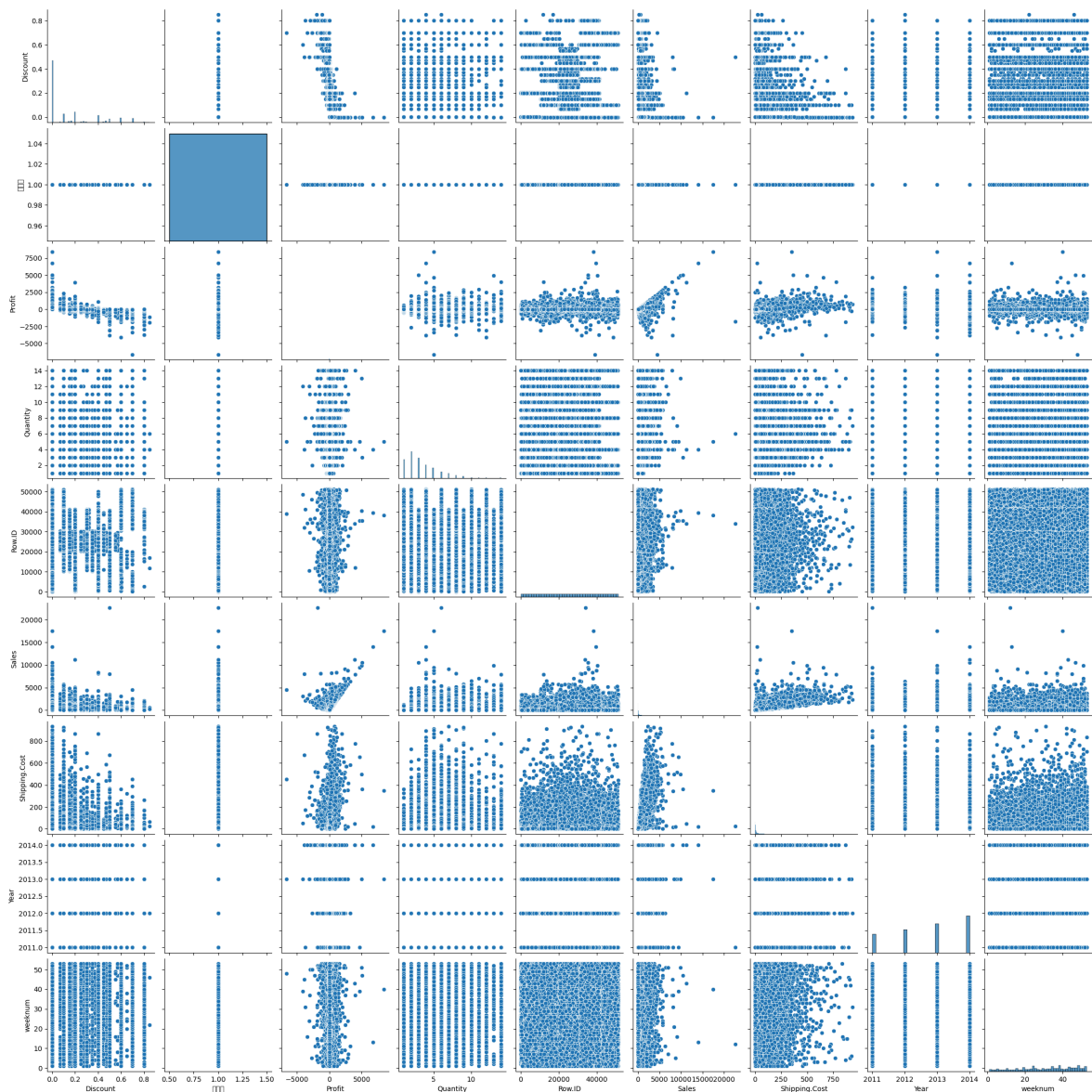
```
C:\ProgramData\anaconda3\Lib\site-packages\seaborn\utils.py:61: UserWarning: Glyph 35760 (\N{CJK UNIFIED IDEOGRAPH-8BB0}) missing from font(s) DejaVu Sans.
  fig.canvas.draw()
C:\ProgramData\anaconda3\Lib\site-packages\seaborn\utils.py:61: UserWarning: Glyph 24405 (\N{CJK UNIFIED IDEOGRAPH-5F55}) missing from font(s) DejaVu Sans.
  fig.canvas.draw()
C:\ProgramData\anaconda3\Lib\site-packages\seaborn\utils.py:61: UserWarning: Glyph 25968 (\N{CJK UNIFIED IDEOGRAPH-6570}) missing from font(s) DejaVu Sans.
  fig.canvas.draw()
C:\ProgramData\anaconda3\Lib\site-packages\IPython\core\pylabtools.py:170: UserWarning: Glyph 35760 (\N{CJK UNIFIED IDEOGRAPH-8BB0}) missing from font(s) DejaVu Sans.
  fig.canvas.print_figure(bytes_io, **kw)
C:\ProgramData\anaconda3\Lib\site-packages\IPython\core\pylabtools.py:170: UserWarning: Glyph 24405 (\N{CJK UNIFIED IDEOGRAPH-5F55}) missing from font(s) DejaVu Sans.
  fig.canvas.print_figure(bytes_io, **kw)
C:\ProgramData\anaconda3\Lib\site-packages\IPython\core\pylabtools.py:170: UserWarning: Glyph 25968 (\N{CJK UNIFIED IDEOGRAPH-6570}) missing from font(s) DejaVu Sans.
  fig.canvas.print_figure(bytes_io, **kw)
```



The heatmap shows the strenght of relationship between numerical variables. Sales and profit appears positively correlated

```
In [10]: sns.pairplot(numeric_df)
plt.show()
```

```
C:\ProgramData\anaconda3\Lib\site-packages\seaborn\axisgrid.py:123: UserWarning:
Glyph 35760 (\N{CJK UNIFIED IDEOGRAPH-8BB0}) missing from font(s) DejaVu Sans.
  self._figure.tight_layout(*args, **kwargs)
C:\ProgramData\anaconda3\Lib\site-packages\seaborn\axisgrid.py:123: UserWarning:
Glyph 24405 (\N{CJK UNIFIED IDEOGRAPH-5F55}) missing from font(s) DejaVu Sans.
  self._figure.tight_layout(*args, **kwargs)
C:\ProgramData\anaconda3\Lib\site-packages\seaborn\axisgrid.py:123: UserWarning:
Glyph 25968 (\N{CJK UNIFIED IDEOGRAPH-6570}) missing from font(s) DejaVu Sans.
  self._figure.tight_layout(*args, **kwargs)
C:\ProgramData\anaconda3\Lib\site-packages\IPython\core\pylabtools.py:170: UserWarning:
Glyph 35760 (\N{CJK UNIFIED IDEOGRAPH-8BB0}) missing from font(s) DejaVu Sans.
  fig.canvas.print_figure(bytes_io, **kw)
C:\ProgramData\anaconda3\Lib\site-packages\IPython\core\pylabtools.py:170: UserWarning:
Glyph 24405 (\N{CJK UNIFIED IDEOGRAPH-5F55}) missing from font(s) DejaVu Sans.
  fig.canvas.print_figure(bytes_io, **kw)
C:\ProgramData\anaconda3\Lib\site-packages\IPython\core\pylabtools.py:170: UserWarning:
Glyph 25968 (\N{CJK UNIFIED IDEOGRAPH-6570}) missing from font(s) DejaVu Sans.
  fig.canvas.print_figure(bytes_io, **kw)
```



The pairplot reveals patterns and relationship among numerical variables and helps identify trends and potential outliers

Summary of Findings

1. Most sales transactions are concentrated at lower values.
2. Several outliers exist in the Sales data.
3. Sales and Profit show a positive relationship.
4. Numerical variables exhibit varying levels of correlation.
5. The dataset contains useful patterns for business decision-making.

In []: